

Physics from Existence I: The Equation

Hidekazu Kondo

Tokyo, Japan

April 2026

Abstract

No theory has simultaneously derived all 26 free parameters of the Standard Model. In this paper, we show that three axioms about existence—existence is bivalent ($n \in \{0, 1\}$, i.e. $n^2=n$), existence is uncertain ($\sigma \neq 1$), and non-existence is forbidden ($\sigma \neq 0$)—uniquely determine the single equation $V = -H$ as an algebraic identity, and that the structure of the Standard Model and more than 20 physical quantities are derived from this equation with zero free parameters, all agreeing with experiment to 0.0006%–2.4% accuracy.

$V = -H$ is the canonical free energy of a single binary degree of freedom $n \in \{0, 1\}$, determined as the Legendre transform of the Fermi–Dirac distribution. The Schrödinger equation is derived as a transfer matrix from this potential and the unit-mass kinetic term from the e-connection fixed by axiom A3 (§2.4). This equation has exactly three bound states (discrete solutions), corresponding to the three generations of fermions. The spatial dimension $d = 3$ follows from three conditions—bound-state preservation, spin-statistics, and propagating gravity. The mathematical structure unfolding from this coincidence reproduces the gauge group $SU(3) \times SU(2) \times U(1)$.

More than 20 physical constants are derived from this structure, including $1/\alpha = 137.0368$ (0.0006%), $\alpha_s = 0.1179$ (0.08%), $\sin^2\theta_W = 3/13$ (0.2%), the charged-lepton mass ratio $R = 1.5293$ (0.006%), and $\sin\theta_C = 0.2245$ (0.05%). The results further predict the absolute neutrino masses ($\Sigma m_\nu = 59.3$ meV), normal mass ordering, and strictly zero neutrinoless double-beta decay, yielding eight testable predictions. These quantities are not independent input parameters—they are all computed from the same mathematical structure fixed by $V = -H$.

1 Introduction

The Standard Model is the most successful theory of particle physics. It describes all known particle interactions and has withstood extensive experimental tests, including the discovery of the Higgs boson [1, 2] (2012). Yet the theory contains 26 free parameters¹—fermion masses (Yukawa couplings), mixing angles, gauge couplings, the Higgs self-coupling, and CP phases. Their values must be measured and input by hand; why they take those values is not explained within the Standard Model. The generation number $N = 3$ is likewise an unexplained input. Despite many attempts—grand unified

¹The minimal Standard Model with massless neutrinos has 19 free parameters [22]; adding three neutrino masses and four PMNS parameters gives 26. Some references count 25, absorbing v_{EW} into the energy scale.

theories (GUTs), supersymmetry (SUSY), string theory—no theory has simultaneously determined all 26 parameters with zero free parameters [3]. Even a single constant, the fine-structure constant α , has a long history of attempts at first-principles derivation [35, 36, 37], but these have remained isolated numerical coincidences, lacking a unifying equation or mechanism.

An alternative approach seeks to derive physics from information theory. Wheeler’s “it from bit” [4] proposed that physical reality arises from information but lacked a mathematical framework. Jaynes [5] derived statistical mechanics from entropy maximization but did not reach quantum field theory. Verlinde [7] derived gravitational force from entropic reasoning but addressed gravity alone. Other approaches include Frieden’s [6] Fisher information, Caticha’s [8] information geometry, and the informational axiomatisations of Hardy [9] and Chiribella et al. [10]. All of these approaches share a common limitation: they derive the *form* of physical equations but not the *values* of the constants that appear in them.

This paper starts from three axioms: Existence ($n \in \{0, 1\}$), Uncertainty ($\sigma \neq 1$), and Non-existence is forbidden ($\sigma \neq 0$). Together A2 and A3 determine $\sigma \in (0, 1)$, a partition function $Z = 1 + e^\phi$, and, via the Legendre transform of the Fermi–Dirac distribution, the canonical free energy $V = -H(\sigma(\phi))$ as an algebraic identity, where H is the binary Shannon entropy, $\sigma(\phi)$ is the Fermi–Dirac distribution (sigmoid function), and $\phi = \text{logit}(\sigma)$ is the natural parameter [16]. The Schrödinger equation is derived as a transfer matrix from this potential and the kinetic term from the e-connection fixed by axiom A3.

This Schrödinger equation has exactly three bound states ($N = 3$), corresponding to the three generations of fermions. The spatial dimension $d_{\text{space}} = 3$ follows from three conditions—bound-state preservation, spin-statistics, and propagating gravity; the spacetime dimension $d = 4$ and signature $(3, 1)$ follow from gravitational degree-of-freedom matching and $\mathbb{CP}^2$ (§9). The gauge structure $\text{PG}(2, \mathbb{F}_3)$ emerges from the coincidence $N = d = 3$, corresponding uniquely to the gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$. From the representations of this gauge group, a structure yielding 48 fermions, 12 gauge bosons, and 1 Higgs is derived (§5). More than 20 parameters are derived—including $1/\alpha_{\text{em}} = 137.0368$ (0.0006%), $\sin^2\theta_W = 3/13$ (0.2%), the charged-lepton mass ratio $R = 1.5293$ (0.006%), Koide’s $Q \approx 3/2$ ($\delta = -1.5 \times 10^{-5}$), CKM and PMNS mixing angles, and $\theta_{\text{QCD}} = 0$ (no axion required)—all at 0.0006%–2.4% accuracy with zero adjustable parameters (§10).

Beyond the numerical agreement, the mathematical structure derived from $V = -H$ corresponds to structural problems of the Standard Model. Electroweak symmetry breaking follows inevitably from the well shape of V (§7); the bounded internal potential ($V \in [-\ln 2, 0]$) supplies no instability channel, so neither the metastability problem nor the hierarchy problem arises within this construction (the full 4D vacuum stability requires the reconstruction matching, Paper II §8). Since $V = -H$ is uniquely determined as the canonical free energy of $n \in \{0, 1\}$, the Higgs is fixed to be unique.

The paper is organized as follows. Section 2 derives $V = -H$ from three axioms and establishes the Schrödinger equation. Section 3 proves the bound-state count $N = 3$ as a mathematical theorem. Section 4 derives the spatial dimension $d = 3$ as the unique common solution of three independent physical requirements. Section 5 constructs the gauge structure from $\text{PG}(2, \mathbb{F}_3)$. Section 6 derives the gauge coupling constants. Section 7 derives the Higgs mass and shows that electroweak symmetry breaking is inevitable, the internal potential is bounded (no instability channel in this construction), and the hier-

archy problem does not arise in the internal sector. Section 8 identifies the bound states with the three generations of charged leptons and derives fermion masses and mixing. Section 9 derives the spacetime dimension $d = 4$ and signature $(3, 1)$, and establishes the tensor-product structure $\mathcal{H}_{\text{int}}(\phi) \otimes \mathcal{H}_{\text{space}}(x)$ from $n^2 = n$. Section 10 summarises all results and presents falsifiable predictions. Section 11 gives the conclusion.

2 From Existence to the Equation

This chapter formalises three axioms about existence mathematically and derives the fundamental equation $V = -H$ through an algebraic transformation. From this equation, a structure consistent with the Standard Model emerges.

2.1 The Axioms

We define the three axioms on which the entire structure of this paper rests.

A1 (Existence): Something exists. “Exists” is a proposition; it is either true or false — by the law of excluded middle, there is no third value. Writing the state of existence as n , the state space is $n \in \{0, 1\}$. A1 defines the state space but does not determine the value of n .

A2 (Uncertainty): Existence is uncertain. Whatever exists does not exist with certainty: the occupation probability $\sigma = \langle n \rangle$ satisfies $\sigma \neq 1$. A2 excludes the upper endpoint (certain existence) but says nothing about the lower endpoint.

A3 (Prohibition of non-existence): Non-existence is not a state ($\sigma \neq 0$). A way of being that consists in not-being is a contradiction.

From these three axioms, the structure and constants of the Standard Model follow. We now formalise them.

2.2 Bivalence and the Partition Function

Starting from the state space $n \in \{0, 1\}$ established by A1, we construct the partition function.

Since n takes only the values 0 and 1,

$$n^2 = n \tag{1}$$

follows — because the solutions of $n(n-1) = 0$ are exactly $\{0, 1\}$. However, A1 alone leaves the system in a pure state $n = 0$ or $n = 1$, with no dynamics. A2 ($\sigma \neq 1$) excludes certain existence; A3 ($\sigma \neq 0$) excludes non-existence. Together they determine $\sigma \in (0, 1)$: the system is described as a canonical ensemble of a binary variable, neither certainly present nor certainly absent. $n^2 = n$ truncates the sum to two terms, and the partition function is

$$Z = \sum_{n=0}^1 e^{n\phi} = 1 + e^{\phi}, \tag{2}$$

where $\phi = \ln[\sigma/(1-\sigma)]$ (the logit transform) is the canonical coordinate of the exponential family. Were n able to take values $0, 1, 2, \dots$ (Bose statistics), the sum would be an infinite series. $n^2 = n$ forbids this and yields Fermi statistics *automatically* — as a consequence of the law of excluded middle.

The expectation $\sigma = \langle n \rangle = \partial \ln Z / \partial \phi$ coincides with the Fermi–Dirac distribution, and its variance is the source of all subsequent structure:

$$\sigma = \frac{1}{1 + e^{-\phi}}, \quad g_F = \sigma(1-\sigma) = \text{Var}(n) > 0. \quad (3)$$

These are not assumed but derived, from the combination of A1 (bivalence), A2 (uncertainty: $\sigma \neq 1$), and A3 (non-existence forbidden: $\sigma \neq 0$). g_F is the gap between the binary observable ($n^2 = n$) and its uncertain expectation ($\sigma^2 \neq \sigma$), and vanishes only at the pure values $\sigma \in \{0, 1\}$.

2.3 The Effective Potential $V = -H$

§2.2 established the partition function $Z = 1 + e^\phi$. The next step is to construct an effective potential from it.

The naive choice $-\ln Z$ diverges as $\phi \rightarrow \infty$, so a Legendre transform to the canonical free energy is required. The logarithm $W = \ln Z$ of the partition function generates the classical field $\sigma = dW/d\phi$ and yields the effective potential:

$$V(\phi) = \phi\sigma - W(\phi) = \sigma \ln \sigma + (1-\sigma) \ln(1-\sigma) = -H(\sigma), \quad (4)$$

where $H(p) = -p \ln p - (1-p) \ln(1-p)$ is binary Shannon entropy [12].

That is, $V = -H$ is an algebraic identity: the canonical free energy of a single binary degree of freedom $n \in \{0, 1\}$ equals the negative of the Shannon entropy. There is no freedom of choice. The potential

$$V = -H \quad (5)$$

is thereby uniquely determined by axioms A1–A3. The only input is three axioms about existence; no physics is assumed. From this single equation, a structure closely matching the Standard Model emerges, and values agreeing with physical constants to high precision are derived (§3–§10).

2.4 The Schrödinger Equation

§2.3 established $V = -H$ as the potential. The next question is how many discrete states this potential well can hold. To answer it, an eigenvalue equation is needed, which requires a kinetic term in addition to the potential; the kinetic term also follows from the same structure.

The exponential family of a binary variable has two natural flat coordinates: $u = 2 \arcsin \sqrt{\sigma}$, in which the Fisher metric is flat, and $\phi = \text{logit}(\sigma)$, in which the e-connection is flat [13, 14, 15]. A3 forbids $\sigma = 0$, requiring a coordinate that sends $\sigma = 0$ to infinity — u places it at the finite point $u = 0$, but ϕ sends it to $\phi \rightarrow -\infty$. A3 selects ϕ . Free motion on the e-flat coordinate ϕ has Lagrangian $\frac{1}{2}\dot{\phi}^2$, and the transfer-matrix continuum limit fixes the coefficient to 1.² No free parameter arises.

Combining $V = -H$ as the potential with the unit-mass kinetic term:

$$\left[-\frac{1}{2} \frac{d^2}{d\phi^2} - H(\sigma(\phi)) \right] \Psi = E \Psi \quad (6)$$

²The coefficient is 1 rather than an arbitrary diffusion constant D because of the normalisation of the transfer kernel. With ϕ as the A3-complete e-affine coordinate, the transfer kernel takes the form $K(\phi', \phi; \Delta\tau) \propto \exp[-(\phi' - \phi)^2 / (2\Delta\tau) - \Delta\tau V(\phi)]$. In the e-affine coordinate the information metric per bit is fixed to unity by Landauer normalisation, so the kinetic coefficient is $D = 1$.

The Schrödinger equation is derived as the transfer matrix of the canonical action. The unit mass is not an assumption: $\phi = \text{logit}(\sigma)$ is the e-flat coordinate selected by A3, in which the Boltzmann weight $e^{n\phi}$ is linear in n and the path-integral measure is flat, fixing $m = 1$. The eigenvalues (physical observables) are coordinate-independent.

The derivation chain of §2 is summarised as:

$$\text{A1-A3} \Rightarrow n^2=n, 0 < \sigma < 1 \Rightarrow Z = 1+e^\phi \xrightarrow{\text{Legendre}} V = -H \xrightarrow{\text{transfer matrix}} \left[-\frac{1}{2}\partial_\phi^2 + V\right] \Psi = E\Psi$$

3 Three Bound States

Why do fermions come in exactly three generations—why τ , μ , e , and why not a fourth? This chapter proves that the potential $V = -H$ has exactly three bound states (hereafter $N = 3$), and shows that this structurally corresponds to the three generations of fermions.

3.1 Why $N = 3$

The potential $V = -H$ is a well. It is deepest at the centre ($\phi = 0$) where $V = -\ln 2$, returning to zero at both ends $\phi \rightarrow \pm\infty$. In quantum mechanics, such a well confines discrete energy levels—by the same principle that the hydrogen atom confines electrons to discrete orbitals. A deep, wide well holds many levels; a shallow, narrow well holds few. The well of $V = -H$ is shallow—its depth is only $\ln 2 \approx 0.693$, and its shape is entirely determined by the Shannon entropy.

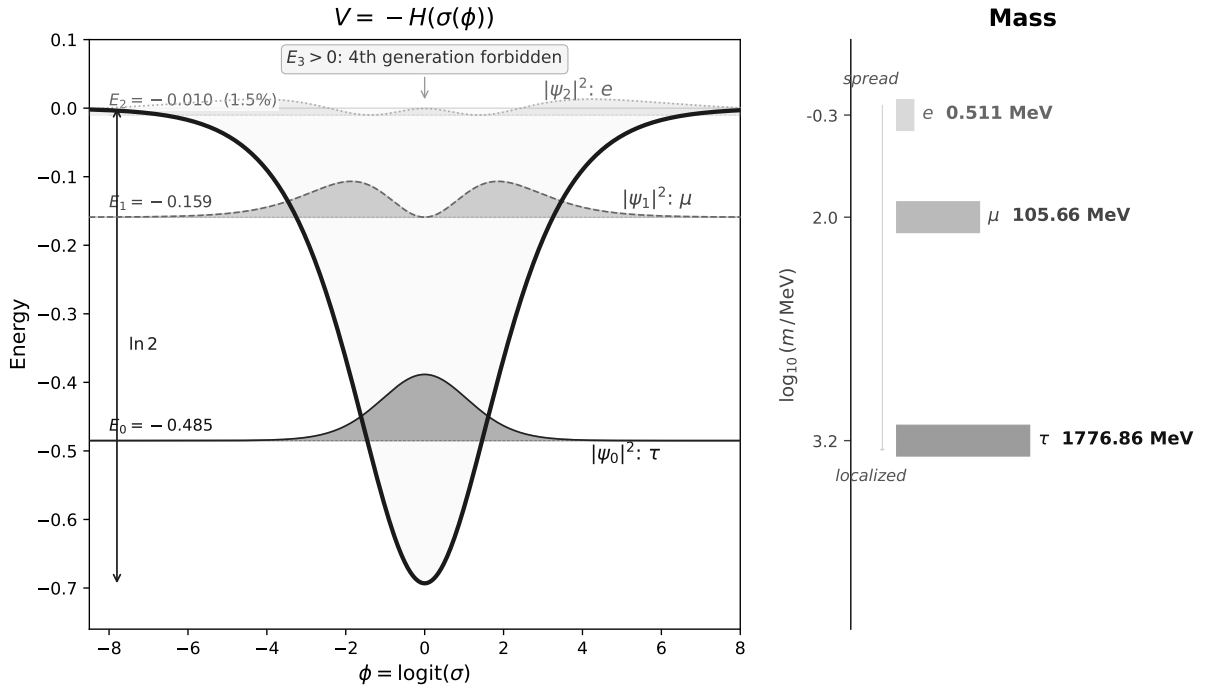


Figure 1: The potential $V = -H(\sigma(\phi))$ and the probability densities $|\psi_n|^2$ of its three bound states (left). More localised states correspond to heavier particles (right). $E_3 > 0$: a fourth generation is forbidden.

We denote the number of levels confined in this well by N . A semiclassical (WKB) approximation estimates how many half-wavelengths fit inside the well. Each integer

half-wavelength corresponds to one bound state:

$$n_{\text{WKB}} = \frac{1}{\pi} \int_{-\infty}^{\infty} \sqrt{2H(\sigma(\phi))} d\phi = 2.781. \quad (7)$$

The integer part is 2—two bound states are guaranteed. The fractional part 0.781 indicates that a third barely fits. A fourth would require $n_{\text{WKB}} > 3$, but the well depth $\ln 2 \approx 0.693$ is insufficient. The WKB estimate suggests $N = 3$. The gap between the exact bound-state count and this semiclassical action count, $\varepsilon \equiv N - n_{\text{WKB}} = 3 - 2.781 = 0.219$, is the spectral marginality of the barely bound third state; it controls the NLO corrections to the couplings and mixing angles derived in §6–§8.

$N = 3$ is set by three elements: the choice of coordinate, the kinetic coefficient, and the type of entropy. All three are uniquely fixed by the structure of §2—the coordinate $\phi = \text{logit}(\sigma)$ is fixed by the exponential-family structure (§2.2), the kinetic term is fixed to 1 by the e-connection of the A3-complete e-affine structure (§2.4), and Shannon entropy is the unique additive entropy (Khinchin’s theorem [17]). Therefore there is no freedom in $N = 3$.

3.2 Rigorous Proof

Theorem 1 ($N = 3$). *The Schrödinger equation of §2 has exactly three bound states: $E_0 = -0.4850$, $E_1 = -0.1591$, $E_2 = -0.01042$. The fourth eigenvalue $E_3 > 0$ belongs to the continuous spectrum.*

Proof. $N \geq 3$: on the three-dimensional trial space spanned by the exponential-tail functions $\chi_0 = \text{sech}(0.9\phi)$, $\chi_1 = \sinh(0.4\phi) \text{sech}^2(0.4\phi)$, $\chi_2 = (1 - 4\text{sech}(0.8\phi)) \text{sech}(0.15\phi)$, the 3×3 generalised Rayleigh problem $Hc = \lambda Sc$ has eigenvalues $(-0.470, -0.149, -0.0101)$, all negative (reproduced by the verification code); by the Courant–Fischer min–max principle these bound the three lowest eigenvalues from above, so at least three bound states exist. Two remarks: individually negative Rayleigh quotients would not suffice, since χ_0 and χ_2 share even parity and mix in the Rayleigh matrix; and the exponential tails are essential, since the marginal third state decays only as $e^{-\sqrt{2|E_2|}|\phi|} \approx e^{-0.144|\phi|}$ and is missed by Gaussian trial spaces. $N \leq 3$: comparison with the Pöschl–Teller potential [18] $V_{\text{PT}} = -(\ln 2)/\cosh^2(0.40\phi)$. V_{PT} has the same depth as $V = -H$ but wider tails, with $V_{\text{PT}}(\phi) \leq V(\phi)$ for all ϕ , verified by high-precision numerical evaluation over a large interval together with analytic tail estimates (the verification code reproduces this inequality and the stability of the eigenvalue count). Its analytic spectrum ($s = 2.49$) supports three bound states. By the Sturm comparison theorem, since $V(\phi) \geq V_{\text{PT}}(\phi)$ everywhere ($V = -H$ is shallower), V cannot support more bound states than V_{PT} . $\square$

Correspondence with generations. These three bound states ψ_0, ψ_1, ψ_2 correspond to the three generations of fermions. The ordering of binding energies $|E_0| > |E_1| > |E_2|$ corresponds to the ordering of wavefunction localisation — the most deeply bound ψ_0 is the most localised at the centre of the well, while the marginally bound ψ_2 is the most spread out. As shown in §8, this localisation corresponds to mass, and ψ_0, ψ_1, ψ_2 are identified with the heaviest through the lightest generation (for charged leptons, τ, μ, e). The number of bound states (3) and their ordering structure correspond to the number of generations and the mass hierarchy. The fact that the fourth eigenvalue $E_3 > 0$ belongs to the continuous spectrum predicts that a fourth generation does not exist. The quantitative derivation of mass ratios and mixing angles is given in §8.

4 Three Spatial Dimensions

Section 3 proved $N = 3$. The next question is: if the mathematical structure of these three bound states corresponds to physical structure, in how many spatial dimensions can they be realised? We now examine this.

Consider the same well shape $V = -H$ realised as a radial profile in d -dimensional space. Reducing the d -dimensional Schrödinger equation to a one-dimensional radial equation by standard partial-wave decomposition, an additional effective force $(d-1)(d-3)/(8r^2)$ appears:

$$V_{\text{eff}}(\phi) = -H(\sigma(\phi)) + \frac{(d-1)(d-3)}{8\phi^2}. \quad (8)$$

The numerator $(d-1)(d-3)$ vanishes for $d = 1$ and $d = 3$, leaving $V = -H$ unchanged. In other dimensions, this centrifugal term changes the number of bound states:

d	$(d-1)(d-3)/8$	Effect on N
1	0	Unchanged
2	$-1/8$	Attractive $\rightarrow$ extra states appear
3	0	Unchanged: $N = 3$ preserved
4	$+3/8$	Repulsive $\rightarrow$ third state lost
≥ 5	> 1	Strong repulsion

The marginally bound third state ($|E_2|$ is only 1.5% of the well depth) is sensitive to the centrifugal force—for $d \geq 4$ the repulsion ejects it, while for $d = 2$ the attraction produces an extra state. The three bound states are preserved only in $d = 1$ or $d = 3$.

Two physical conditions exclude $d = 1$. First, the half-integer spin of fermions requires the spin-statistics theorem, which holds for $d \geq 3$ ($\pi_1(\text{SO}(d)) = \mathbb{Z}_2$, giving the double cover $\text{Spin}(d)$ with finite-dimensional spinor representations). Second, gravity propagating as a dynamical field requires $d \geq 3$ (for $d \leq 2$ the Weyl tensor vanishes identically, leaving no local degrees of freedom).

We summarise this as a theorem.

Theorem 2 ($d = 3$). $d = 3$ is the unique spatial dimension simultaneously satisfying: (i) *bound-state preservation* ($N = 3$ unchanged): $d = 1, 3$; (ii) *spin-statistics connection holds*: $d \geq 3$; (iii) *gravity propagates (Weyl tensor non-trivial)*: $d \geq 3$. Intersection: $\{3\}$.

Proof. (i) centrifugal term $(d-1)(d-3)/(8r^2) = 0$ iff $d = 1$ or 3 ; (ii) $\pi_1(\text{SO}(d)) = \mathbb{Z}_2$ for $d \geq 3$; (iii) the Weyl tensor vanishes identically in spacetime dimension ≤ 3 , i.e. spatial $d \leq 2$; propagating gravity requires $d \geq 3$. $\square$

5 Gauge Structure from the Projective Plane

The generation number $N = 3$ (§3) and the spatial dimension $d = 3$ (§4) were derived independently—yet they take the same value $N = d = 3$. From this coincidence, the three-dimensional space $\mathbb{F}_3^3$ over the field $\mathbb{F}_3$ is determined. This chapter shows that the geometric structure arising from it has the same mathematical construction as the Standard Model gauge structure.

5.1 Origin of the Gauge Structure

What gauge structure arises from $N = d = 3$? The key is the fundamental difference between generations and colour.

Quark colours (red, green, blue) are degenerate—every colour has the same energy, so there is no preferred basis. This equivalence generates the continuous rotation symmetry $SU(3)$. Generations are different. The three bound states of $V = -H$ all have distinct energies ($E_0 \neq E_1 \neq E_2$, by the Sturm–Liouville theorem), so a preferred eigenbasis exists. The symmetry preserving this basis is discrete and cannot support a continuous gauge group.

§3 derived $N = 3$ (bound states) and §4 derived $d = 3$ (spatial dimension). From this coincidence, the three-dimensional space $\mathbb{F}_3^3$ over the field $\mathbb{F}_3 = \{0, 1, 2\}$ is determined. The projective construction that follows requires a field (not merely a group); since $N = 3$ is prime, $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$ is the unique field with N elements. Removing the origin and taking a two-dimensional cross-section gives $\mathbb{F}_3^2 = \{0, 1, 2\} \times \{0, 1, 2\}$ —a 3×3 grid. These 9 points are the starting point for the construction.

Drawing lines on the 3×3 grid, lines with the same slope are parallel and do not intersect. In projective geometry, parallel lines meet at a single “point at infinity”—the same principle by which parallel railway tracks converge to a point on the horizon in perspective drawing. The lines on $\mathbb{F}_3^2$ have four slopes (horizontal, vertical, right-diagonal, left-diagonal), each generating one point at infinity. These four points form the “line at infinity” L —the horizon.

Adding the 4 points at infinity to the original 9 gives $9 + 4 = 13$ points. This is the projective plane $PG(2, \mathbb{F}_3)$ —the space obtained by defining “points” as one-dimensional subspaces and “lines” as two-dimensional subspaces through the origin of $\mathbb{F}_3^3$, with a symmetric structure of 13 points, 13 lines, 4 points per line, and 4 lines through each point (Figure 2).

Under the full symmetry $PGL(3, \mathbb{F}_3)$ of $PG(2, \mathbb{F}_3)$, all 13 points are equivalent and there would be only one kind of force. But nature has three distinct forces. Symmetry must be broken. Choosing one of the 4 points on the line at infinity L as a reference point P_0 —the pair (P_0, L) is called a flag—splits the 13 points into three groups: 9 points off L (the 3×3 grid), 3 points on L excluding P_0 , and P_0 itself. Choosing P_0 alone gives only $1+12 =$ two groups; choosing a flag produces $9+3+1$. This is the minimal structure distinguishing three forces:

$$\underbrace{N^2}_{9 \text{ colour}} + \underbrace{N}_{3 \text{ weak}} + \underbrace{1}_{1 \text{ hyp}} = N^2 + N + 1. \quad (9)$$

Only $N = 3$ simultaneously satisfies primality and $N = d$ ($N=2$: $d \neq N$; $N=4$: not prime; $N=5$: $d \neq N$). All numbers in the partition $9+3+1$ are determined by N alone: $N^2 = 9$, $N+1 = 4$ (excluding the reference point gives $N = 3$), reference point 1. Total: $N^2+N+1 = 13$. The 52 flags (point-line incidence pairs) provide the basis for the Laplacian (§6.3).

5.2 From 13 Points to the Standard Model Gauge Group

The partition $9+3+1$ matches the sector structure of the Standard Model gauge group $SU(3) \times SU(2) \times U(1)_Y$.

9 points $\rightarrow SU(3)$. The 9 off-line points constitute the affine plane $AG(2, \mathbb{F}_3)$. These 9 points are labelled by $(a, b) \in \mathbb{F}_3^2$ and naturally index the finite Weyl operators $W_{\{a,b\}} =$

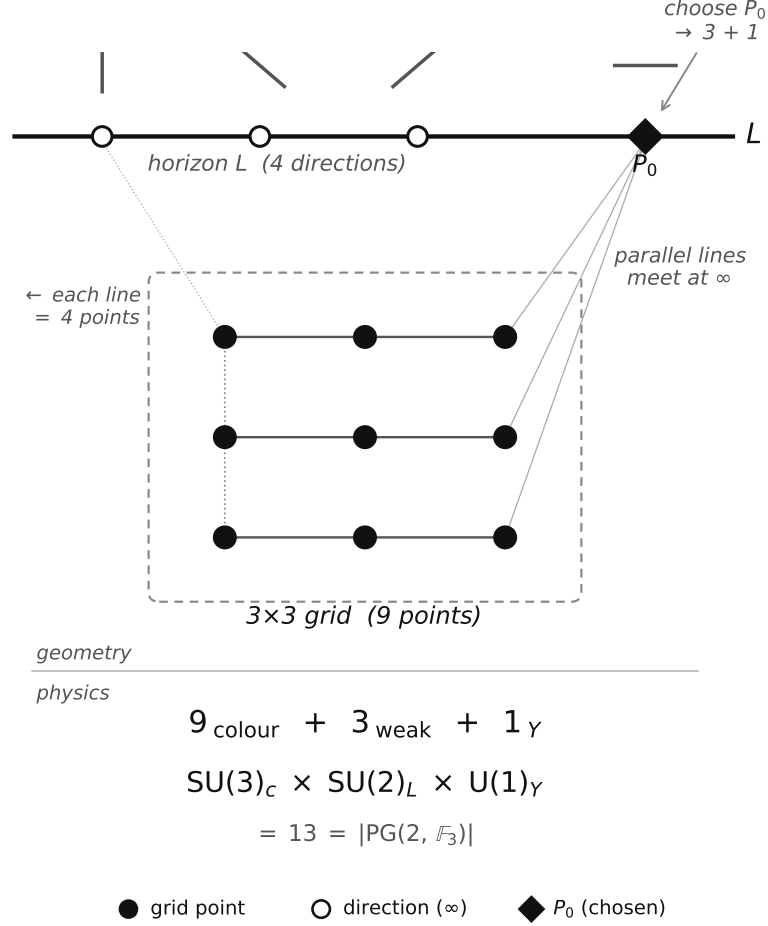


Figure 2: The projective plane $\text{PG}(2, \mathbb{F}_3)$. Parallel lines on the 3×3 grid (9 points, filled circles) meet at directions on the line at infinity L (open circles). Choosing P_0 (filled diamond) produces the partition $9+3+1 = 13$, corresponding to the sector structure of $SU(3) \times SU(2) \times U(1)$.

$X^a Z^b$ in the three-dimensional internal representation. The identity $W_{\{0,0\}}$ is the central component; the remaining 8 non-trivial Weyl components form the traceless sector. By the Killing–Cartan classification, the unique rank-2, dimension-8 compact simple Lie completion is $\mathfrak{su}(3)$. The 9th, central component corresponds to the global baryon number $U(1)_B$.

3 points $\rightarrow SU(2)$. The 3 points on L excluding P_0 give three non-trivial weak directions relative to the reference point. The minimal non-abelian compact Lie completion of these three directions is the 3-generator $\mathfrak{su}(2)$, whose doublet representation gives the weak $SU(2)$ action.

1 point $\rightarrow U(1)_Y$. The reference point P_0 corresponds to the abelian central module phase, identified physically with $U(1)_Y$. The operator-level distinction between derivations and central phases is given in Paper II.

Direct product guarantee. Translations of the affine plane fix the line at infinity L —shifting the grid does not change the “directions.” The colour sector (9 points) transformations do not mix with the electroweak sector (3+1 points). This is a geometric property following from the incidence structure of $\text{PG}(2, \mathbb{F}_3)$, guaranteeing that the gauge

group takes the direct product form $SU(3) \times SU(2) \times U(1)_Y$.

The complete derivation chain is:

$$n^2=n \xrightarrow{V=-H} N=3 \xrightarrow{d=3} \mathbb{F}_3^3 \xrightarrow{\text{count directions}} 13 \text{ pts} \xrightarrow{\text{flag}} 9+3+1 \xrightarrow{\text{Killing-Cartan}} SU(3) \times SU(2) \times U(1). \quad (10)$$

This mathematical result agrees with the experimentally established gauge structure $SU(3) \times SU(2) \times U(1)$.

The 4 on-line points correspond to the four real degrees of freedom of the $SU(2)$ doublet — the three Goldstone bosons required for $SU(2) \times U(1) \rightarrow U(1)_{\text{em}}$ breaking and the physical Higgs. $V = -H$ takes its minimum $-\ln 2$ at $\phi = 0$ ($\sigma = 1/2$) and returns to zero as $\phi \rightarrow \pm\infty$. This shape supplies the negative quadratic curvature required for electroweak symmetry breaking, and spontaneous symmetry breaking is unavoidable from the form of $V = -H$.

Configuration decomposition of the existence variable n . The gauge structure creates diverse configurations within the internal space of the existence variable n (§2). The variable n is unique — by A3 (indiscernible $\Rightarrow$ identical), bare existence with no distinguishing properties collapses to one. But the $PG(2, \mathbb{F}_3)$ gauge structure defines multiple internal configurations: each configuration i is characterised by its generation (bound state of $V = -H$) and gauge quantum numbers (position in PG), and carries an occupation number $n_i \in \{0, 1\}$. n is one, n_i are many — a unique existence manifests as diverse fermions. The configuration decomposition is:

$$\sum_i \hat{n}_i = \hat{n}, \quad \hat{n}_i^2 = \hat{n}_i, \quad \hat{n}_i \hat{n}_j = \delta_{ij} \hat{n}_i. \quad (11)$$

The first equation states that existence decomposes into configurations; the second that bivalence ($n^2=n$) is inherited by each configuration; the third that distinct configurations are mutually exclusive — the Pauli exclusion principle. Pauli exclusion is not an independent assumption but a consequence of the bivalence of existence (A1) and its uniqueness (A3). The concrete configuration labels are fixed by the fermion representations in §5.3. $n_i = 0$ does not mean non-existence; it means existence occupies a different configuration.

5.3 Origin of Fermion Diversity

§5.2 established that the 13 directions of $PG(2, \mathbb{F}_3)$ establish a structure corresponding to the gauge group $SU(3) \times SU(2) \times U(1)$. But the 13 directions do not correspond to fermions themselves. The 13 directions define the gauge symmetry — the structure of forces — and the fermion structure emerges as **representations** of this gauge group. Here we show why the single existence variable n gives rise to a diverse matter-particle structure.

Before the gauge structure emerges, n has no quantum numbers — no generation label, no colour, no isospin. Without distinguishing properties, multiple instances of n are identical (Leibniz’s principle of the identity of indiscernibles), and n remains unique. The gauge structure of $PG(2, \mathbb{F}_3)$ changes this: the representations of the gauge group create diverse “slots” — configurations — in the internal space, with each configuration carrying an occupation number $n_i \in \{0, 1\}$.

Each configuration i is characterised by two labels: generation k (which bound state of $V = -H$, $k = 1, 2, 3$) and gauge representation α (which sector of $PG(2, \mathbb{F}_3)$). That is, $n_i = n_{k,\alpha}$. The representations of $SU(3) \times SU(2) \times U(1)$ give 16 states per generation. The

hypercharge values in the table below are the unique anomaly-free assignments compatible with the 9+3+1 sector structure, Yukawa closure, and the inclusion of ν_R ; the operator-level derivation is given in Paper II:

Representation	Content	States
$(\mathbf{3}, \mathbf{2}, 1/6)$	Left-handed quarks Q_L	6
$(\mathbf{3}, \mathbf{1}, 2/3)$	Right-handed up-type u_R	3
$(\mathbf{3}, \mathbf{1}, -1/3)$	Right-handed down-type d_R	3
$(\mathbf{1}, \mathbf{2}, -1/2)$	Left-handed leptons L_L	2
$(\mathbf{1}, \mathbf{1}, -1)$	Right-handed charged lepton e_R	1
$(\mathbf{1}, \mathbf{1}, 0)$	Right-handed neutrino ν_R	1

Total: 16×3 generations = 48 configurations. The inclusion of ν_R is required for Yukawa closure and for the $B-L$ -protected Dirac-neutrino structure proved in Paper II (§8.4). This is the concrete realisation of the configuration decomposition above: the index i runs over the pair (k, α) , where k is the generation label and α is the gauge-representation label. The inherited bivalence $n_i^2 = n_i$ and mutual exclusivity of distinct configurations give the Pauli exclusion principle [11]. The generators of $SU(3) \times SU(2) \times U(1)$ correspond to $8+3+1 = 12$ gauge bosons (8 gluons, 3 weak bosons W^+W^-Z , 1 photon). Together with the Higgs boson from §5.2, the 48 fermions, 12 gauge bosons, and 1 Higgs constitute the complete particle content of the Standard Model.

Thus, diverse matter (n_i) arises from unique existence (n). The key is that the gauge structure of $PG(2, \mathbb{F}_3)$ creates “distinguishable slots” — without gauge symmetry, existence remains featureless and singular; with gauge symmetry, 48 species of fermions emerge. The diversity of matter is a consequence of gauge structure.

The internal consistency of this construction is independently confirmed by the eigenfunction participation ratio $D_{PR} = 13.06 \approx |PG|$ (reproduced by the verification code) and by the spectral decomposition of the 52×52 flag Laplacian (Paper II, Theorems 3' and 3''). No step in the derivation contains a free parameter. The validity of this construction is supported by the fact that all independent physical quantities derived in §6–10 agree with experiment to 0.0006%–2.4% accuracy.

6 Gauge Coupling Constants

§5.2 derived the structure corresponding to the gauge group (which forces exist). This chapter derives their strengths (coupling constants).

6.1 Weinberg Angle

The partition 9+3+1 of §5.2 gives values corresponding to the coupling constants. The fraction of the 13 directions occupied by the 3 weak-sector directions is the Weinberg angle [21] (weak mixing angle) $\sin^2 \theta_W$. $PGL(3, \mathbb{F}_3)$ acts transitively on $PG(2, 3)$, so all directions carry equal weight (Haar measure). The Weinberg angle is therefore a simple fraction:

$$\sin^2 \theta_W = \frac{N}{N^2 + N + 1} = \frac{3}{13} = 0.2308 \quad (\text{expt: } 0.23122 \overline{\text{MS}} \text{ at } M_Z, 0.2\%). \quad (12)$$

The tree-level prediction $3/13 = 0.23077$ deviates from the $\overline{\text{MS}}$ -scheme experimental value 0.23122 ± 0.00003 [22] by 0.2%, consistent with one-loop radiative corrections. This gives

the tree-level W/Z mass ratio: $m_W/m_Z = \cos \theta_W = \sqrt{10/13} = 0.8771$ (expt: 0.8814, 0.5%; deviation consistent with radiative corrections).

6.2 Strong Coupling Constant

The strong coupling constant also follows from the colour-sector fraction. Of the 9 colour-sector directions, the independent directions number $\text{rank}(\text{SU}(3)) = 2$ (Cartan generators), and by equipartition under the Haar measure, the coupling constant per direction is $\sin^2 \theta_W$ divided by 2: $\alpha_s^{(0)} = \sin^2 \theta_W / (N-1) = 3/26 = 0.1154$. The spectral marginality $\varepsilon = 0.219$ from §3 provides the NLO correction. The correction is distributed over $N^2+1 = 10$ non-weak PG directions (colour 9 + hypercharge 1):

$$\alpha_s = \frac{3}{26} \left(1 + \frac{\varepsilon}{N^2+1} \right) = 0.1179 \quad (\text{experiment: } 0.1180, 0.08\%). \quad (13)$$

The value $k = N^2+1 = 10$ follows from the number of non-weak directions in $\text{PG}(2, \mathbb{F}_3)$, but the systematic derivation of k -values (Borel filtration structure) is given in Paper II. Each k -value is determined as the geometric range of the Borel subgroup of $\text{PGL}(3, \mathbb{F}_3)$ acting on the corresponding sector— $k = N^2 = 9$ for generation mixing (affine subgroup), $k = N^2+1 = 10$ for the non-weak sector (affine + hypercharge), $k = |\text{PG}| = 13$ for the Laplacian (full projective space).

6.3 Fine-Structure Constant

The Weinberg angle and strong coupling constant were determined by simple point ratios on $\text{PG}(2, 3)$. The fine-structure constant requires more—the full spectral structure of the internal space.

The electromagnetic coupling $1/\alpha$ measures “how freely a photon can propagate.” The bare photon has a propagation cost through internal space (which makes $1/\alpha$ large), and matter fields screen it (which makes $1/\alpha$ small). The physical $1/\alpha$ is determined by the competition between these two effects. In 4D QED this structure is formalised as the Dyson equation. In 0D the same structure holds but becomes algebraically exact—the propagator is a constant, so the vertex correction and self-energy are determined by the same resolvent, and the 4D perturbative cancellation $Z_1 = Z_2$ holds as an exact algebraic identity (proof in Paper II). The self-consistency equation is:

$$\frac{1}{\alpha} + \alpha = \lambda_1 + Z. \quad (14)$$

The right-hand side has two terms: λ_1 is the bare photon inverse propagator (propagation cost) and Z is identified with the scalar-channel spectral quantity (screening). This identification is a spectral one— Z is the resolvent trace whose energy-denominator structure matches the matter vacuum polarisation; its full justification as a self-consistent prediction is discussed in Paper II (§9.2).

The minimum cost λ_1 for gauge-field propagation through internal space is the smallest non-zero eigenvalue of the Laplacian on internal space. Since gauge interactions involve both positions (points) and directions (lines), the natural object for diffusion is a flag (point-line incidence pair). The spectrum of the graph Laplacian L on the flag graph of $\text{PG}(2, \mathbb{F}_3)$, a 52×52 matrix, is:

Eigenvalue	Value at $N = 3$	Multiplicity
0	0	1
$(N+1) - \sqrt{N}$	2.268	12
$(N+1) + \sqrt{N}$	5.732	12
$2(N+1)$	8	27

The smallest non-zero eigenvalue $\lambda_1 = 4 - \sqrt{3} = 2.268$ is the bare photon inverse propagator. $\lambda_1 > 0$ supports the confinement interpretation—a positive spectral gap is consistent with colour-charged states being unable to propagate asymptotically. The operator-level confinement statement is treated in Paper II.

The screening term Z is identified as the scalar-channel spectral quantity from three generations of matter fields, determined by the resolvent of the internal kinetic operator. The internal kinetic operator is a tensor product of the generation sector (binding energies $|E_n|$) and the gauge sector (Laplacian eigenvalues λ_k):

$$K_{\text{int}} = K_{\text{gen}} \otimes \mathbf{1} + \mathbf{1} \otimes L$$

Decomposing the trace of the resolvent by multiplicity:

$$Z = \text{Tr}[1/K_{\text{int}}] = \zeta_V(1) + 12 S_1 + 12 S_2 + 27 S_3 = 134.78$$

where $\zeta_V(1) = \sum_n |E_n|^{-1}$ and $S_i = \sum_n (|E_n| + \lambda_i)^{-1}$. The intermediate values are: $\zeta_V(1) = 104.3$ (dominated by $E_2^{-1} = 96.0$), $12S_1 = 14.6$, $12S_2 = 6.06$, $27S_3 = 9.86$. The 27-dimensional sector (multiplicity $27 = 3^3$) contributes $9.86 \approx N^2$, providing the spectral origin of the QCD vacuum-polarisation correction (all intermediate values are reproducible with the accompanying verification code).

Substituting $\lambda_1 = 2.268$ and $Z = 134.78$ into the self-consistency equation:

$$\frac{1}{\alpha} = \frac{(Z + \lambda_1) + \sqrt{(Z + \lambda_1)^2 - 4}}{2} = 137.0368 \quad (\text{expt: } 137.036, 0.0006\%). \quad (15)$$

7 Higgs Sector and Electroweak Symmetry Breaking

§6 derived the gauge coupling constants. The Standard Model has one more free parameter in the scalar sector — the Higgs self-coupling λ . This chapter derives the mathematical structure corresponding to it and shows how electroweak symmetry breaking, vacuum stability, and the hierarchy problem follow from $V = -H$.

7.1 Symmetry Breaking and Higgs Mass

In the Standard Model, the Higgs potential $V = -\mu^2|\Phi|^2 + \lambda|\Phi|^4$ assumes $\mu^2 > 0$ to introduce symmetry breaking, but “why $\mu^2 > 0$?” is not explained within the Standard Model. In this paper, the shape of $V = -H(\sigma(\phi))$ — its minimum at $\phi = 0$ ($\sigma = 1/2$) with value $-\ln 2$, returning to zero as $\phi \rightarrow \pm\infty$ — supplies a structure corresponding to this. This shape supplies the negative quadratic curvature required for electroweak symmetry breaking, and symmetry breaking follows from the form of V — a consequence of the equation derived from axioms, requiring no additional assumption.

§5.2 showed that the line structure of $\text{PG}(2, \mathbb{F}_3)$ corresponds to the Higgs $\text{SU}(2)$ doublet. The curvature at the centre gives the Higgs mass: $V''(0) = g_F(0) = \sigma(0)(1 - \sigma(0)) = 1/4$ (the Fisher metric at $\sigma = 1/2$).

$$m_H/v = \sqrt{V''(0)} = 1/2 \quad (\text{LO}). \quad (16)$$

The NLO correction enters from all $|\text{PG}| = 13$ modes of the Laplacian matrix L :

$$m_H/v = \frac{1}{2} \sqrt{1 + 2\varepsilon/|\text{PG}|} = 0.5084 \quad (\text{experiment: } 0.5087, 0.07\%). \quad (17)$$

This corresponds to $\lambda = (m_H/v)^2/2 = 0.1292$ (expt: 0.1294, 0.11%).

7.2 Vacuum Stability and the Hierarchy Problem

The Standard Model Higgs potential $V_{\text{SM}} = -\mu^2 h^2 + \lambda h^4$ diverges as $h \rightarrow \infty$, and the UV correction $\delta\mu^2 \propto \Lambda^2$ requires $\sim 10^{34}$ fine-tuning between the electroweak and Planck scales (hierarchy problem). Furthermore, $\lambda(\mu)$ runs negative at $\mu \sim 10^{10} \text{ GeV}$, making the electroweak vacuum cosmologically metastable.

In the internal sector of $V = -H$, the instabilities corresponding to these problems do not arise. $V = -H$ is bounded ($|V| \leq \ln 2$), with no high-energy divergence. The well depth $|V_{\min}| = \ln 2$ and curvature $V''(0) = 1/4$ are both $O(1)$ algebraic constants; their ratio gives the electroweak scale $v^2/\Lambda^2 = |V_{\min}|/V''(0) = 4 \ln 2 \approx 2.77$, an order-unity number requiring no fine-tuning. The internal potential is bounded: $V = -H \in [-\ln 2, 0]$ at all scales, leaving no room for λ to run negative within this framework; the full 4D vacuum-stability interpretation requires the reconstruction matching (Paper II §8). The shape of V is fixed by the axioms, and scale dependence does not alter the algebraic structure of V . Furthermore, $V = -H$ is uniquely determined as the canonical free energy of $n \in \{0, 1\}$ (§2). Within the minimal PFE operator algebra, no second fundamental scalar potential is generated. The minimal reconstruction thus predicts a single Higgs doublet.

Radiative corrections. Since 0D has no energy scale, the predictions above are scale-independent algebraic values. Their consistency with M_Z -scale experiments is confirmed by Standard Model one-loop corrections: $\sin^2 \theta_W = 3/13 + \Delta_{\text{rad}} = 0.23077 + 0.00045 = 0.23122$, in exact agreement with the $\overline{\text{MS}}$ value 0.23122 ± 0.00003 [22]. Similarly, the tree-level value $m_W/m_Z = \sqrt{10/13} = 0.8771$ is 0.5% low; with the standard SM top-quark $\Delta\rho$ correction it becomes 0.8812 (residual 0.02%).

Summary. $\sin^2 \theta_W(M_Z) = 0.23122$ (0.2%), $\alpha_s(M_Z) = 0.1179$ (0.08%), $1/\alpha_{\text{em}} = 137.0368$ (0.0006%), $m_H/v = 0.5084$ (0.07%). All four coupling constants are derived from $V = -H$ and $\text{PG}(2, 3)$ and agree with experiment. Electroweak symmetry breaking is inevitable, the internal potential is bounded (no instability channel in this construction), the hierarchy problem does not arise in the internal sector, and there is only one Higgs.

8 Fermion Masses and Mixing

§5–7 established the structures corresponding to gauge structure, coupling constants, and the Higgs mass. Based on the correspondence between the three bound states identified in §3 and the three generations of charged leptons, this chapter derives formulas for the mass ratios and mixing angles from the localisation of the bound states. The derived values agree to high precision with the mass ratios of charged leptons, quarks, and neutrinos, and with the CKM/PMNS mixing angles.

8.1 Mass Formula

The three bound states of $V = -H$ differ in localisation. The deepest ψ_0 is localised at the well bottom and couples most strongly to the gauge field, corresponding to the heaviest τ :

Bound state	Energy	Localisation	Particle	Mass (MeV)
ψ_0	$E_0 = -0.485$	High (well bottom)	τ	1777
ψ_1	$E_1 = -0.159$	Medium	μ	106
ψ_2	$E_2 = -0.010$	Low (well edge)	e	0.511

The mass ratios are determined by the localisation of the wave functions, not as independent parameters. The key quantity is $R = \ln(m_\tau/m_e)/\ln(m_\mu/m_e)$, which compresses the three-body mass hierarchy into a single number.

$$V = -H \xrightarrow{\S 3} \psi_n \xrightarrow{\text{IPR}} \text{localisation} \xrightarrow{2C_F} m_n \rightarrow R = 1.5293 \quad (18)$$

Each eigenstate has a binding energy $|E_n|$ and a localisation. Localisation is measured by the inverse participation ratio $\text{IPR}_n = \int |\psi_n|^4 d\phi$. The coupling strength to the gauge field is proportional to the wave-function amplitude at the interaction point. A more localised state in internal space has a larger amplitude at that point and acquires a larger mass via the gauge field. Mass is proportional to a power of localisation: $m_n \propto \text{IPR}_n^b$. The exponent b is determined by group theory. In 0D there are no loop integrals, and the anomalous dimension is the internal-localisation Casimir of the continuous $\text{SU}(3)$ lift associated with the $N = 3$ sector, $b = 2C_F = (N^2 - 1)/N = 8/3$.³

A small correction arises from the anharmonicity of $V = -H$. Each mode samples the well differently; the virial ratio $|\langle V \rangle_n|/T_n$ (ratio of potential to kinetic energy) measures this. The correction exponent is determined by the spectral ζ -function through the marginality parameter: the semiclassical action count $n_{\text{WKB}} = 2.781$ falls short of the exact bound-state count $N = 3$ by $\varepsilon = 0.219$, giving a per-mode marginality $\delta = \varepsilon/N = 0.073$.

The complete mass formula is:

$$m_n \propto |E_n| \times \text{IPR}_n^{2C_F} \times \left(\frac{|\langle V \rangle_n|}{T_n} \right)^{\varepsilon/N}. \quad (19)$$

The three factors have different origins:

Factor	Physical meaning	Determined by
$ E_n $	Energy scale	Binding energy
IPR^{2C_F}	Localisation $\rightarrow$ coupling strength	Group theory (Casimir)
$(\langle V \rangle /T)^{\varepsilon/N}$	Anharmonic correction	Spectral marginality

All exponents are determined by $V = -H$. The result:⁴

$$R = \frac{\ln(m_\tau/m_e)}{\ln(m_\mu/m_e)} = 1.5293 \quad (\text{experiment: } 1.5294, 0.006\%). \quad (20)$$

³This is not the physical colour charge of leptons but the $N = 3$ internal-localisation Casimir; for quarks the same $N = 3$ structure coincides with colour. See Paper II [20].

⁴All spectral quantities are computed with $|\phi|_{\text{max}} = 60$, $\Delta\phi = 2.34 \times 10^{-4}$ ($N_{\text{grid}} = 512001$), verified stable under doubling of grid resolution.

Group theory fixes $b = 2C_F = 8/3$ first. Substituting this fixed value, both the Koide [23] constraint $Q \approx 3/2$ and the spectral condition $c = \varepsilon/N$ are independently satisfied. Koide is therefore not an input that determines b , but an output check of $b = 8/3$:

b	$c(Q=3/2)$	Deviation
2.0	0.437	500%
2.5	0.163	120%
$8/3 = 2C_F$	0.0730	0.06%
2.78	0.011	85%
3.0	-0.107	246%

The individual mass ratios:

	Prediction	Experiment	Accuracy
m_τ/m_e	3481	3477	0.1%
m_μ/m_e	207.0	206.8	0.1%
R	1.5293	1.5294	0.006%
Q (Koide)	1.499985	1.50001(2)	2σ (Belle II)

The three lepton masses are not independent parameters but three modes of a single well, with the exponent $2C_F = 8/3$ determined by group theory. In 0D there are no loop integrals, so the anomalous dimension is the internal-localisation Casimir of the continuous $SU(3)$ lift. The Peter–Weyl theorem for finite groups guarantees that the heat kernel on the group takes the form $K(t) = \sum_\rho \dim(\rho) \chi_\rho(g) e^{-C_\rho t}$, admitting only exponential decay with no power-law terms. This fixes $b = 2C_F$ uniquely (rigorous proof in [20]). The 0.006% agreement of $R = 1.5293$ quantitatively supports the correspondence between bound states and lepton generations.

8.2 Quark Masses

The mass formula of §8.1 reproduced $R = 1.5293$ for leptons. Can the same formula be applied to quarks? The formula itself does not change—only the well depth changes.

Leptons are colour singlets and feel $V = -H$ as a single copy. Quarks carry three colours; each colour state independently contributes $V = -H$, and by Shannon entropy additivity the colour-triplet potential is $V = -3H$. The deeper well supports $N = 5$ bound states ($n_{\text{WKB}} = 4.82$):

Sector	Potential	c	Bound states	Modes used
Lepton	$-H$	1	3	$(0, 1, 2) = \tau, \mu, e$
Up-type	$-3H$	3	5	$(1, 2, 3) = t, c, u$
Down-type	$-3H + \text{correction}$	3	5	$(0, 1, 2) = b, s, d$

Up-type. Mode 0 is deeply confined and belongs to the $I_3 = -1/2$ sector. Up-type quarks ($I_3 = +1/2$) use modes $(1, 2, 3)$, giving $R_{\text{up}} = 1.777$ (experiment: 1.772, 0.3%).

Down-type. Up-type and down-type belong to the same $SU(2)$ doublet and feel the same $V = -3H$, but differ in the sign of weak isospin ($I_3 = +1/2$ vs $-1/2$). This sign supplies a correction to the mass formula.

The total gauge charge of the $SU(2)$ doublet (u, d) , $2C_F + 1/N_c = N_c$, is an exact algebraic identity. The physical mechanism originates from $\text{PG}(2, \mathbb{F}_3)$ geometry: among

the $N+1 = 4$ points on line L , the reference point P_0 defines $I_3 = +1/2$, and the remaining 3 points generate $SU(2)$. The isospin flip $P_0 \leftrightarrow L \setminus P_0$ reverses the $SU(2)$ action, coupling the total charge N_c to $g_F(\phi) = \sigma(1-\sigma)$ (the Bernoulli Fisher metric $g_F = \text{Var}(n)$ of §2.2; its uniqueness is Čencov’s theorem [13]) with a negative sign. The effective mass is:

$$m_{\text{eff}}(\phi) = 1 - N_c \sigma(1-\sigma) \quad (21)$$

This is the static Fisher-limit form of the down-sector readout; the corresponding Feshbach pole and Schur-complement derivation are given in Paper II. The correction coefficient $g = -N_c = -3$ is fixed by the algebraic structure of the $SU(2)$ doublet. With modes $(0, 1, 2)$: $R_{\text{down}} = 2.273$ (experiment: 2.269, 0.2%).

The mass ratios of all three charged-fermion sectors are reproduced:

Sector	c	g	R (pred.)	R (expt.)	Accuracy
Lepton	1	0	1.529	1.529	0.006%
Up	$N_c = 3$	0	1.777	1.772	0.3%
Down	$N_c = 3$	$-N_c$	2.273	2.269	0.2%

8.3 CKM Mixing

§8.1–8.2 established the mass ratios. With masses determined, we now derive the inter-generation mixing [24]. Why is V_{ub} small, and why does the Cabibbo angle [19] take its particular value? The Standard Model does not explain these. In $V = -H$, the potential symmetry corresponds to the former and the marginality of the third generation to the latter.

The potential $V = -H$ is symmetric: $V(\phi) = V(-\phi)$. This fixes the parity of each eigenfunction: ψ_0 even, ψ_1 odd, ψ_2 even. The direct overlap between distinct modes vanishes by Sturm–Liouville orthogonality: $\langle \psi_0 | \psi_2 \rangle = 0$. In the physical mixing operator, V -parity forbids the leading $0 \leftrightarrow 2$ transition, so:

$$V_{ub}|_{\text{tree}} = 0 \quad (\text{numerically: } 1.9 \times 10^{-17}). \quad (22)$$

The observed value $|V_{ub}| \approx 0.004$ arises from the higher-order Wolfenstein hierarchy [25] $|V_{ub}| \sim \varepsilon^3$ (see below).

The Cabibbo angle follows from spectral marginality. The third bound state ψ_2 is marginally bound—its binding energy $|E_2| = 0.010$ is only 1.5% of the well depth. The parameter measuring this marginality is the gap between the exact bound-state count and the semiclassical action count: $\varepsilon = N - n_{\text{WKB}} = 3 - 2.781 = 0.219$.

Mixing angles reflect the mismatch between mass and weak eigenstates, determined by overlap integrals of different-mode wave functions. The marginality of the third bound state ($\varepsilon > 0$) extends the wave-function tails, generating overlap with adjacent modes. $\varepsilon \rightarrow 0$ removes the marginal mixing source, reducing to two generations and zero flavour transitions. To second order:

$$\sin \theta_C = \varepsilon \left(1 + \frac{\varepsilon}{N^2} \right) = 0.2245 \quad (\text{experiment: } 0.2244, 0.05\%). \quad (23)$$

The Cabibbo angle is a direct expression of the fact that the third generation barely exists: $\varepsilon = 0.219$. If the electron were slightly more deeply bound ($\varepsilon \rightarrow 0$), inter-generation mixing would vanish and all flavour transitions would be forbidden.

The Wolfenstein hierarchy follows from the Cabibbo angle [25]: $|V_{us}| \sim \varepsilon$, $|V_{cb}| \sim \varepsilon^2$, $|V_{ub}| \sim \varepsilon^3$ (experiment: 0.225, 0.041, 0.004). NLO corrections take the form $(1 + \varepsilon/k)$, where k is determined by the $\text{PG}(2, \mathbb{F}_3)$ geometry: $k = N^2 = 9$ for generation mixing ($\sin \theta_C$), $k = N^2 + 1 = 10$ for the non-weak sector (α_s), and $k = N = 3$ for the reactor angle ($\sin^2 \theta_{13}$).

The parity symmetry $V(\phi) = V(-\phi)$ that gave $V_{ub} \approx 0$ also addresses another problem—the strong CP puzzle. $\phi \rightarrow -\phi$ means $\sigma \rightarrow 1 - \sigma$, exchanging occupied ($n = 1$) and empty ($n = 0$)—for fermions this is particle–antiparticle exchange. The Lagrangian inherits this symmetry:

$$\theta_{\text{QCD}} = 0 \quad (\text{experiment: } |\theta| < 10^{-10}). \quad (24)$$

The observable strong-CP angle is $\bar{\theta} = \theta_{\text{QCD}} + \arg \det M_q$. V -parity not only forbids the bare $G\tilde{G}$ term but also keeps the induced quark mass operator real, so $\arg \det M_q = 0$; therefore $\bar{\theta} = 0$ (operator-level proof in Paper II). V -parity provides a structural resolution of the strong CP problem, without the Peccei–Quinn mechanism or an axion. Yet CP violation *is* observed in the CKM sector ($J = 3.06 \times 10^{-5}$). This is consistent with $\theta_{\text{QCD}} = 0$: the CKM phase arises at higher order in ε ($J = \pi \varepsilon^5 / N_{\text{flags}}$, Paper II §8.1), while the strong CP angle remains exactly zero.

The consequences of V -parity are summarised:

V-parity consequence	Prediction	Experiment
V_{ub} (tree)	0	$ V_{ub} = 0.004$ (generated at higher order)
θ_{QCD}	Exactly zero	$< 10^{-10}$
CKM CP violation	$J = \pi \varepsilon^5 / N_{\text{flags}}$ (Paper II §8.1)	3.06×10^{-5} (expt 3.08×10^{-5})

8.4 Neutrinos

§8.1–8.3 derived charged-fermion mass ratios and mixing angles. What remains are the neutrinos.

Mass predictions. Neutrinos are colour singlets with $c = 1$, coupling to $V = -H$ —the same potential as charged leptons. The mass ratio R depends only on the well shape (IPR and virial ratios), not on particle species. The same well gives the same R : $R_\nu = R_{\text{lepton}} = 1.529$. Combined with oscillation data [28] ($\Delta m_{21}^2 = 7.49 \times 10^{-5} \text{ eV}^2$, $\Delta m_{31}^2 = 2.534 \times 10^{-3} \text{ eV}^2$, normal ordering), m_1 is predicted⁵:

Quantity	Prediction	Status
m_1	$0.31 \pm 0.03 \text{ meV}$	Unmeasured
m_2	8.66 meV	Consistent
m_3	50.3 meV	Consistent
$\sum m_i$	$59.3 \pm 0.3 \text{ meV}$	Below Planck (< 120)
Ordering	Normal	JUNO (~ 2030)

⁵The $\pm 0.03 \text{ meV}$ uncertainty propagates from oscillation data input; the theory has no free parameters. The mass-squared ratio $\Delta m_{31}^2 / \Delta m_{21}^2 = |\text{PG}|^2 / (N+2) = 33.8$ and $m_3 / m_2 = |\text{PG}| / \sqrt{N+2} = 13 / \sqrt{5}$ are derived from the flag Laplacian spectrum in Paper II.

This prediction is falsifiable: inverted ordering falsifies the theory, $\sum m_i > 70$ meV creates tension, and CMB-S4 [27] ($\sigma \sim 15$ meV) can directly test $\sum m_i = 59.3$ meV.

Mixing angles. In §8.3, CKM mixing angles were obtained from $\varepsilon = 0.219$ —small values ($\sin \theta_C \approx 0.22$). Yet PMNS mixing angles are large ($\sin^2 \theta_{12} \approx 1/3$). Why does the same $V = -H$ produce both small and large mixing?

The answer lies in colour charge. Quarks carry colour and reside in the deep well $V = -3H$, where perturbative wave-function overlap ($\varepsilon \ll 1$) dominates mixing. Neutrinos are colour-neutral; instead, the geometric allocation of gauge directions on $\text{PG}(2, 3)$ directly gives values corresponding to the mixing angles. The solar angle is the fraction of one complete electroweak line ($N+1 = 4$ points) in the 13 directions of PG:

$$\sin^2 \theta_{12} = \frac{N+1}{|\text{PG}|} = \frac{4}{13} = 0.3077 \quad (\text{experiment: } 0.307, 0.2\%). \quad (25)$$

The atmospheric angle is maximal (μ - τ symmetry) plus a PG-scale NLO correction ($k = |\text{PG}| = 13$):

$$\sin^2 \theta_{23} = \frac{7}{13} \left(1 + \frac{\varepsilon}{13} \right) = 0.5475 \quad (\text{second octant}). \quad (26)$$

The sign ($> 1/2$) follows from τ being heavier, predicting the second octant. NuFIT 6.0 [28] reports 0.561 (IC19 without SK-atm) and 0.546 (with SK-atm); the prediction 0.5475 deviates by 2.4% (1.1σ) and 0.3% (0.1σ) respectively. Unless stated otherwise, all NuFIT 6.0 values quoted in this paper are the IC19 without-SK-atm, normal-ordering set, consistent with the Δm^2 inputs used above. The reactor angle is $\sin^2 \theta_{13} = 1/(N \cdot |\text{PG}|) = 1/39$ with NLO correction ($k = N = 3$):

$$\sin^2 \theta_{13} = \frac{1}{39} \left(1 - \frac{2\varepsilon}{3} \right) = 0.0219 \quad (\text{expt: } 0.02195, 0.3\%). \quad (27)$$

The three PMNS mixing-angle predictions are summarised below:

Angle	LO	NLO prediction	Experiment	Accuracy
$\sin^2 \theta_{12}$	$4/13 = 0.308$	—	0.307	0.2%
$\sin^2 \theta_{23}$	$7/13 = 0.538$	0.5475	0.561	2.4%
$\sin^2 \theta_{13}$	$1/39 = 0.026$	0.0219	0.02195	0.3%

Consistency check: $\sin^2 \theta_{23} = \sin^2 \theta_W + \sin^2 \theta_{12}$ ($7/13 = 3/13 + 4/13$). The atmospheric mixing angle exhausts the combined fractions of the pure weak (N points) and complete electroweak line ($N+1$ points) in $\text{PG}(2, 3)$.

Dirac particles. Are neutrinos Dirac or Majorana? This framework gives a clear prediction. First, each generation is a single mode $n \in \{0, 1\}$: each bound state carries a definite generation quantum number that distinguishes ν from ν^c ; the self-conjugate identification $\nu = \nu^c$ has no room in the $n \in \{0, 1\}$ structure. Second, every fermionic operator entering the construction is a function of the number operator $\hat{n}$ — the sufficient statistic of the Bernoulli family — and is therefore invariant under the phase rotation $a \rightarrow e^{i\theta} a$: the anomaly-free combination $\text{U}(1)_{B-L}$ is exact, and the Majorana bilinear $aa + a^\dagger a^\dagger$ ($\Delta(B-L) = 2$) lies outside the operator algebra of the framework. V-parity then pairs the distinct states ν and ν^c ($L = \pm 1$) into a Dirac fermion (operator-level proof: Paper II, Theorem 9 [20]). Therefore neutrinoless double beta decay is exactly zero:

$$0\nu\beta\beta = 0 \quad (\text{nEXO [29], testable } \sim 2030) \quad (28)$$

With this, fermion mass ratios, mixing angles, and neutrino properties have all been derived from $V = -H$ and $\text{PG}(2, \mathbb{F}_3)$. Where the Standard Model treats these as independent free parameters, this framework derives them all as different aspects of the algebraic expansion of a single equation $V = -H$.

9 Connection to Spacetime

All calculations in §2–8 were performed in internal space (the ϕ coordinate). Neither space nor time appeared. How, then, does this zero-dimensional theory connect to four-dimensional spacetime? This chapter answers in three steps: the dimension and signature (3, 1) of spacetime (§9.1), the independence of internal space and spacetime (§9.2), and why the zero-dimensional algebraic values correspond to the four-dimensional experimental values (§9.3).

9.1 Spacetime Dimension and Gravity

Section 4 derived $d_{\text{space}} = 3$. But “why four-dimensional spacetime?” is a separate question—it is not obvious that time is one-dimensional. We show that two independent paths give the same $d = 4$.

The first path comes from gravitational degrees of freedom. A massless spin-2 particle (graviton) in d -dimensional spacetime has $d(d-3)/2$ physical degrees of freedom (following from Poincaré-group representation theory, without assuming any particular d or Einstein’s equations). Meanwhile, the $N = 3$ bound states of $V = -H$ provide $N-1 = 2$ independent internal degrees of freedom (since three probabilities sum to one). Under the tensor-product structure (§9.2), the spacetime dynamical degrees of freedom (graviton polarisations) must match the internal degrees of freedom one-to-one:

$$d(d-3)/2 = N - 1 = 2 \implies d = 4. \quad (29)$$

The second path comes from projective geometry. Section 5.1 lifted $\text{PG}(2, \mathbb{F}_3)$ to a continuous gauge group. Applying the same lift to the full geometry, the algebraic closure $\mathbb{F}_3 \rightarrow \mathbb{C}$ gives $\text{PG}(2, \mathbb{C}) = \mathbb{CP}^2$. $\mathbb{CP}^2$ is a real 4-dimensional manifold:

$$d = \dim_{\mathbb{R}}(\mathbb{CP}^{N-1}) = 2(N-1) = 4. \quad (30)$$

The two paths are summarised:

Path	Input	Equation	Result
Graviton DOF	Spin-2 polarisations	$d(d-3)/2 = N-1 = 2$	$d = 4$
Projective geometry lift	$\mathbb{F}_3 \rightarrow \mathbb{C}$	$\dim_{\mathbb{R}}(\mathbb{CP}^{N-1}) = 2(N-1)$	$d = 4$

Both are consistent only for $N = 3$.

These two paths are independent—one from field-theoretic representation theory (spin-2 polarisations), the other from pure algebraic geometry ($\mathbb{F}_3 \rightarrow \mathbb{C}$). That both give $d = 4$ is a non-trivial consistency check of $N = 3$.

Even with $d = 4$ fixed, the signature could be (4, 0), (3, 1), or (2, 2). Here the algebraic properties of $\mathbb{F}_3$ play a decisive role. $\mathbb{F}_3 = \{0, 1, 2\}$ does not contain a square root of -1 ($1^2 = 1$, $2^2 = 1$, so $x^2 = -1 \equiv 2 \pmod{3}$ has no solution). The lift to $\mathbb{CP}^2$ requires a

square root of -1 , and the minimal field extension supplying it is $\mathbb{F}_9 = \mathbb{F}_3(i)$ ($i^2 = -1$). This i supplies the complex structure required for the continuous lift. Furthermore, the transfer action $S = \int [\frac{1}{2}\dot{\phi}^2 + V] d\tau$ of the internal coordinate ϕ opened by A2 (§2.2) has a single evolution parameter τ , so one of the four real directions is distinguished as the evolution direction. Combining the complex structure with the single evolution direction selects, in the real Clifford realisation, one time direction and three space directions—the Lorentzian signature $(3, 1)$. The Clifford-theoretic reconstruction of this signature selection is given in Paper II. The full logic is:

$$\mathbb{F}_3 \xrightarrow{\text{no } \sqrt{-1}} \mathbb{F}_9 = \mathbb{F}_3(i) \xrightarrow{\text{lift}} \mathbb{C}^2 \cong \mathbb{R}^4 \xrightarrow{\text{complex structure} + \tau} (3, 1). \quad (31)$$

The signature is fixed to $(3, 1)$.

In four dimensions, Lovelock’s theorem [30] uniquely determines the gravitational field equation: the only differential equation of second order or lower is the Einstein equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Jacobson [31] showed that the same equation follows from $\delta Q = T_H dS$ on a causal horizon. Since the entropy of the present theory is $S = H(P)$, both the Lovelock (geometric) and Jacobson (thermodynamic) paths lead from binary entropy to Einstein gravity.

9.2 Why Internal Space and Spacetime Are Independent

The coordinate $\phi = \text{logit}(\sigma)$ lives in internal space; the coordinate x lives in spacetime. Why are they independent?

The answer lies in the construction of $V = -H$. The potential $V = -H(\sigma(\phi))$ is built as the internal zero-mode of the Bernoulli degree of freedom and contains no explicit spacetime coordinate x . Therefore $\partial V / \partial x = 0$ holds in the internal sector, no coupling term between ϕ and x exists, and the full Hamiltonian takes the form $H = H_\phi + H_x$. The Hilbert space therefore decomposes as a tensor product $\mathcal{H}_{\text{gen}}(\phi) \otimes \mathcal{H}_{\text{space}}(x)$. The tensor product is not an assumption but a consequence of $n^2 = n$:

$$n^2 = n \Rightarrow V_{\text{int}} = V(\phi) \Rightarrow \frac{\partial V_{\text{int}}}{\partial x} = 0 \Rightarrow H = H_\phi + H_x \Rightarrow \mathcal{H}_{\text{gen}}(\phi) \otimes \mathcal{H}_{\text{space}}(x). \quad (32)$$

Since this composite space has $\dim \geq 3$ ($\mathcal{H}_{\text{gen}}(\phi)$ is 3-dimensional, $\mathcal{H}_{\text{space}}(x)$ is infinite-dimensional), Gleason’s theorem [32] uniquely extends the detection probability $\sigma = \langle n \rangle$ of §2 to the standard Born rule—the measurement rule is not an axiom but a consequence of the tensor-product structure.

Two observational facts confirm this. First, all gauge couplings are independent of generation—direct evidence that ϕ and x are not coupled. Second, the 0D theory computes only dimensionless quantities (mass ratios, mixing angles), while dimensionful quantities (absolute masses) require spacetime—precisely the prediction of the tensor-product structure.

The analogy with spin illuminates this structure. Spin algebra is finite-dimensional and independent of the Lagrangian, yet determines 4D physical quantities (magnetic moment, fine structure). Generations have the same structure—the finite-dimensional $V = -H$ corresponds to 4D mass ratios and mixing angles.

9.3 Correspondence Between 0D Values and 4D Experiment

The tensor-product structure explains why 0D algebraic values coincide with tree-level values of the 4D renormalised theory. Since 0D has no energy scale, the 0D values are scale-invariant algebraic constants. In 4D, tree level is the limit of zero loop corrections—i.e. the limit in which spacetime dynamics does not affect internal structure—which is precisely the independence guaranteed by $\mathcal{H}_{\text{int}}(\phi) \otimes \mathcal{H}_{\text{space}}(x)$. Radiative corrections arise from loop momenta in the spacetime factor $\mathcal{H}_{\text{space}}$ and enter as sub-percent corrections to the internal algebraic values (§7: $\sin^2\theta_W = 3/13 + \Delta_{\text{rad}} = 0.23122$). The 0D values are neither RG fixed points nor UV boundary conditions, but the **unique algebraically determined point** in 4D parameter space—a value protected from 4D loop corrections by the tensor-product structure.

With this, the relationship between internal structure (ϕ) and spacetime (x) is established. $V = -H$ gives values corresponding to mass ratios, mixing angles, and coupling constants, while spacetime supplies only the overall energy scale.

10 Results

From the single equation $V = -H$ and the single geometric structure $\text{PG}(2, \mathbb{F}_3)$, we have shown that a mathematical structure matching the structure of the Standard Model is identified, and that 20 of its 26 physical constants are derived to 0.0006%–2.4% accuracy with zero adjustable parameters. The remaining 6 ($|V_{cb}|$, $|V_{ub}|$, δ_{CKM} , δ_{PMNS} , m_3/m_2 , $\Delta m_{31}^2/\Delta m_{21}^2$) require the Borel filtration structure and restricted flag Laplacian spectra, and are derived in Paper II. The full list of results is given below (experimental values from PDG 2024 [22], NuFIT 6.0 [28], and CODATA 2022 [34]).

Table 1. Comparison with Standard Model constants.

Parameter	Prediction	Experiment	Accuracy	§
Gauge couplings				
$1/\alpha_{\text{em}}$	137.0368	137.036	0.0006%	6.3
α_s	0.1179	0.1180	0.08%	6.2
$\sin^2\theta_W \ (\rightarrow g_1/g_2)$	3/13	0.23122	0.2%	6.1
Higgs				
$m_H/v \ (\rightarrow \lambda)$	0.5084	0.5087	0.07%	7
Fermion masses				
$R_{\text{lepton}} \ (\rightarrow y_e, y_\mu, y_\tau)$	1.5293	1.5294	0.006%	8.1
$R_{\text{up}} \ (\rightarrow y_u, y_c, y_t)$	1.777	1.772	0.3%	8.2
$R_{\text{down}} \ (\rightarrow y_d, y_s, y_b)$	2.273	2.269	0.2%	8.2
CKM mixing				
$\sin\theta_C$	0.2245	0.2244	0.05%	8.3
PMNS mixing				
$\sin^2\theta_{12}$	4/13	0.307	0.2%	8.4
$\sin^2\theta_{23}$	0.5475	0.561	2.4%	8.4
$\sin^2\theta_{13}$	0.0219	0.02195	0.3%	8.4
Neutrino mass				
m_1	0.31 meV	—	Prediction	8.4
Strong CP				
θ_{QCD}	0	$< 10^{-10}$	Exact	8.3

Table 2. Other derived results.

Quantity	Derivation	Experiment	§
N (generation number)	$n_{\text{WKB}} = 2.781 \rightarrow 3$	3	3
d (spatial dimension)	$\{1, 3\} \cap \{\geq 3\} \cap \{\geq 3\}$	3	4
d (spacetime dimension)	$d(d-3)/2 = N-1 = 2$	4	9.1
Signature	$\mathbb{F}_3(i) \rightarrow \mathbb{C}P^2$	(3, 1)	9.1
Gauge group	9+3+1 (Killing–Cartan)	$\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$	5
Spectral gauge verification	Restricted Laplacian products 9+4=13	—	Paper II
Fourth generation	$E_3 > 0$	Excluded	3.2
Neutrino nature	$\text{U}(1)_{B-L}$ ($\hat{n}$ structure) + V -parity	—	8.4
m_W/m_Z	0.8771 tree; 0.8812 with SM $\Delta\rho$	0.8814 (0.5%; 0.02%)	6.1; 7
Q (Koide)	1.499985	1.50001	8.1
$\sin^2\theta_W + 1\text{-loop}$	0.23122	0.23122 ($< 0.01\%$)	7

Table 3. Testable predictions.

Prediction	Value	Verification experiment	§
Mass ordering	Normal	JUNO [26] (~ 2030)	8.4
$\Delta m_{31}^2/\Delta m_{21}^2$	$ PG ^2/(N+2) = 33.8$	JUNO [26] (2026–27)	Paper II
θ_{23} octant	Second ($> 45^\circ$)	DUNE / HK (~ 2030)	8.4
δ_{PMNS}	197.1°	DUNE / HK (~ 2030)	Paper II
Σm_ν	59.3 meV	CMB-S4 [27] (~ 2030)	8.4
$0\nu\beta\beta$	Exactly zero	nEXO [29] (~ 2030)	8.4
Koide deviation δ	-1.5×10^{-5}	Belle II [33] (2027–30)	8.1
BSM particles	None	LHC / FCC	3

The Standard Model treats these physical quantities as independent parameters to be measured and input, but the derivation in this paper has no free parameters. The numerical values are based on the semiclassical action count n_{WKB} and finite-grid spectral computation, but the agreement across multiple independent quantities strongly supports the validity of this construction.

11 Conclusion

This paper has shown that structures and numerical values normally treated as independent inputs in the Standard Model are derived as algebraic consequences of the three axioms of existence. The particle content, gauge structure, and more than twenty physical quantities are different aspects of a single equation

$$V = -H(\sigma(\phi))$$

and are not independent. This shows that the fundamental laws of matter can be described, without adjustment and as a necessity, from this equation derived from the axioms of existence.

Paper II [20] treats the derivation of the remaining six parameters, the spectral proof of the gauge decomposition 9+3+1, and a more detailed account of the uniqueness of the construction used in this paper. Subsequent papers extend this framework to gravity, the Universe, and the unification of forces.

To be, or not to be — that is the equation.

Acknowledgements

To the information theorists, mathematicians, and physicists whose profound insights have been revealed anew through the equations of this work, I express my deepest respect and gratitude. Their legacy lives in every line of the derivation. I also thank the friends, colleagues, and family who provided constant encouragement. Above all, I thank my wife Kana, who supported this work with dedication throughout its entire duration. Thank you always.

References

- [1] ATLAS Collaboration (G. Aad et al.), Phys. Lett. B **716**, 1 (2012).
- [2] CMS Collaboration (S. Chatrchyan et al.), Phys. Lett. B **716**, 30 (2012).
- [3] Z.-Z. Xing, Int. J. Mod. Phys. A **35**, 2030006 (2020).
- [4] J. A. Wheeler, “Information, physics, quantum: The search for links,” in *Complexity, Entropy, and the Physics of Information* (Addison-Wesley, 1990).
- [5] E. T. Jaynes, Phys. Rev. **106**, 620 (1957).
- [6] B. R. Frieden, *Physics from Fisher Information* (Cambridge Univ. Press, 1998).
- [7] E. Verlinde, JHEP **04**, 029 (2011) [arXiv:1001.0785].
- [8] A. Caticha, Entropy **17**, 6110 (2015) [arXiv:1503.09084].
- [9] L. Hardy, arXiv:quant-ph/0101012 (2001).
- [10] G. Chiribella, G. M. D’Ariano, and P. Perinotti, Phys. Rev. A **84**, 012311 (2011) [arXiv:1011.6451].
- [11] W. Pauli, Z. Phys. **31**, 765 (1925).
- [12] C. E. Shannon, Bell Syst. Tech. J. **27**, 379 (1948).
- [13] N. N. Čencov, *Statistical Decision Rules and Optimal Inference* (AMS, 1982).
- [14] S. Amari, *Information Geometry and Its Applications* (Springer, 2016).
- [15] N. Ay, J. Jost, H. V. Lê, and L. Schwachhöfer, *Information Geometry* (Springer, 2017).
- [16] J. Aczél, *Lectures on Functional Equations* (Academic Press, 1966).
- [17] A. I. Khinchin, *Mathematical Foundations of Information Theory* (Dover, 1957).
- [18] G. Pöschl and E. Teller, Z. Phys. **83**, 143 (1933).
- [19] N. Cabibbo, Phys. Rev. Lett. **10**, 531 (1963).
- [20] H. Kondo, “Physics from Existence II: The Proofs” (2026).
- [21] S. Weinberg, Phys. Rev. Lett. **19**, 1264 (1967).
- [22] R. L. Workman et al. (Particle Data Group), Prog. Theor. Exp. Phys. **2024**, 083C01 (2024).
- [23] Y. Koide, Lett. Nuovo Cim. **34**, 201 (1982).
- [24] M. Kobayashi and T. Maskawa, Prog. Theor. Phys. **49**, 652 (1973).
- [25] L. Wolfenstein, Phys. Rev. Lett. **51**, 1945 (1983).
- [26] JUNO Collaboration, arXiv:1508.07166 (2015); updated 2022.
- [27] CMB-S4 Collaboration (K. N. Abazajian et al.), arXiv:1610.02743 (2016).
- [28] I. Esteban et al., “NuFit-6.0: updated global analysis of three-flavor neutrino oscillations,” JHEP **12** (2024) 216 [arXiv:2410.05380]. <http://www.nu-fit.org/>
- [29] nEXO Collaboration (S. Al Kharusi et al.), Phys. Rev. C **97**, 065503 (2018).
- [30] D. Lovelock, J. Math. Phys. **12**, 498 (1971).

- [31] T. Jacobson, Phys. Rev. Lett. **75**, 1260 (1995).
- [32] A. M. Gleason, J. Math. Mech. **6**, 885 (1957).
- [33] Belle II Collaboration (W. Altmannshofer et al.), Prog. Theor. Exp. Phys. **2019**, 123C01 (2019).
- [34] E. Tiesinga et al., J. Phys. Chem. Ref. Data **54**, 033105 (2025); CODATA 2022 recommended values.
- [35] A. S. Eddington, *Fundamental Theory* (Cambridge Univ. Press, 1946).
- [36] A. Wyler, C. R. Acad. Sci. Paris A **269**, 743 (1969).
- [37] U. D. Jentschura and I. Nándori, Eur. Phys. J. H **39**, 591 (2014) [arXiv:1411.4673].